LINCC Frameworks

LSDB

Universidad de Antioquia, Open Tools, Open sky

Scientific goal and workflow

Building a variable-star sample from public time-domain catalogs

This notebook demonstrates a reproducible workflow for identifying known variable stars in Rubin alert data and enriching them with astrometric and photometric information from Gaia DR3.

Workflow

- Select known variable stars from the VSX catalog.
- Perform a positional crossmatch with Rubin alert sources.
- Recover the available alert and historical photometric measurements.
- Crossmatch the resulting sample with Gaia DR3.
- Inspect and export the combined scientific dataset.

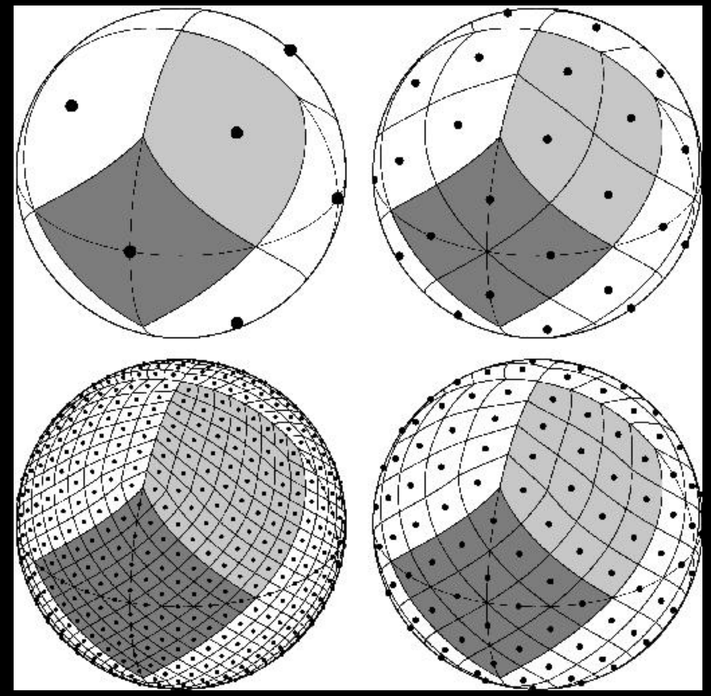


Why LSDB and HATS?

Scalable analysis of large astronomical catalogs

Large surveys cannot always be handled efficiently as conventional in-memory tables. LSDB operates on catalogs organized in the HATS format, in which the sky is divided into spatial partitions.

- Only the relevant sky partitions are accessed during a query.
- Catalogs can be inspected and filtered before the full computation is executed.
- Positional crossmatches are performed directly on spatially partitioned data.
- Rubin detection histories can remain associated with their parent sources as nested time-series information.
- The same workflow can be tested on a small region and subsequently scaled to a much larger sample.

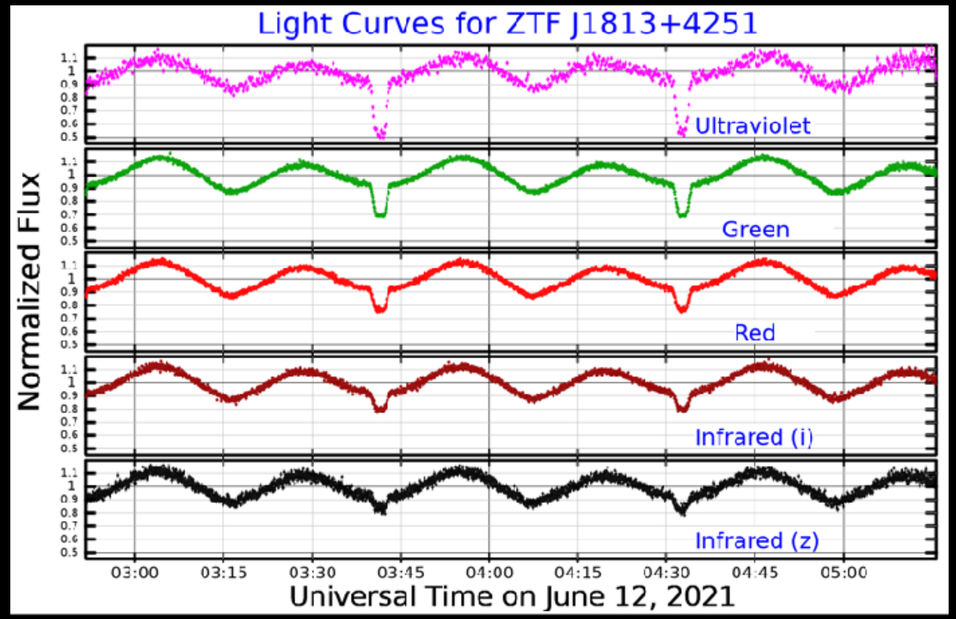


From catalog matching to scientific analysis

Scientific products generated by the notebook

For each matched variable-star candidate, the workflow can provide:

- The original VSX identifier and variability classification
- Rubin alert detections and historical photometric measurements
- Multiband light-curve information
- Gaia DR3 counterpart and angular match separation
- Parallax, proper motion, mean photometry, and quality indicators
- Compact table that can be saved and reused in later analyses



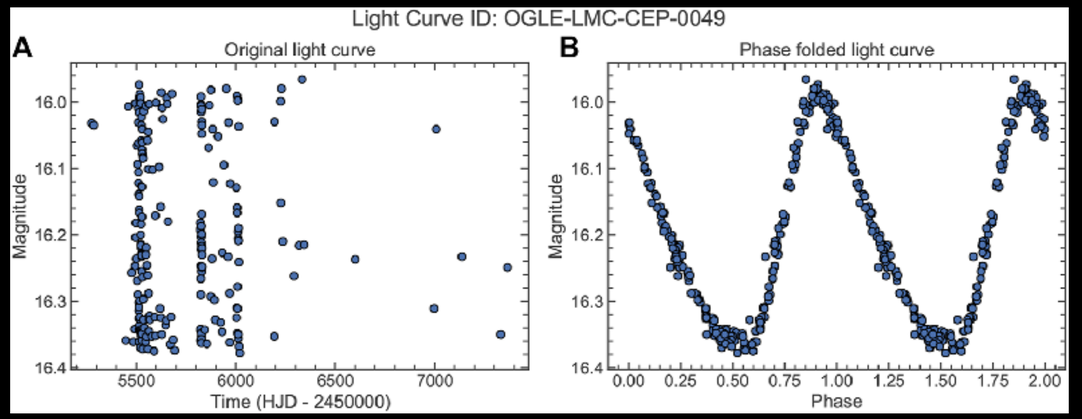
From catalog matching to scientific analysis

Scientific products generated by the notebook

These products can support:

- visual inspection and validation of variable-star classifications
- period searches and phased light-curve analysis
- color–magnitude and kinematic studies
- identification of unusual or potentially misclassified objects
- preparation of science-ready samples for subsequent tutorials or research projects.

The final result is not simply a catalog crossmatch, but a traceable and reusable dataset for time-domain astrophysics.



Let’s see the notebook



Scan me!

https://colab.research.google.com/drive/1xZqspogrGtA_kSDJP6j_xf9B3XNqlloc?usp=sharing

How LSDB Organizes Large Sky Catalogs

HEALPix + HATS + Parquet + Dask

- **HEALPix** divides the celestial sphere into hierarchical, equal-area pixels.
- **HATS** adaptively partitions the catalog according to source density.
- Dense regions use smaller pixels; sparse regions use larger pixels.
- Each spatial partition is stored as a **Parquet file**.
- **Dask** enables lazy, parallel, and distributed computation.
- LSDB reads only the partitions and columns required by the query.
- **Margin catalogs** preserve sources near pixel boundaries during crossmatching.
- Spatial partitioning makes large-scale catalog queries and crossmatches efficient